

A Voyage to the Kernel



Part 19

Day 18: Segment 3.8

*A*s I promised in the last issue, in this article, we will explore networking.

OSI reference model

The OSI (Open System Interconnection) reference model is a multi-layered computer network protocol architecture that has seven distinct layers from top to bottom (refer to Figure 1). These are the application, presentation, session, transport, network, data and physical layers. After covering some of the unique features of the architecture, we will move on to networking in Linux systems.

The whole purpose of this division is to subtly divide the process of networking. Conceptually, the layer below a particular layer is therefore one layer to provide support to the former. By definition, you can have a 'link' between two instances belonging to a particular type of layer. You can assume that there is a kind of *horizontal protocol* connection existing between them.

Let's discuss these layers from the bottom to the top. The physical layer, as the name suggests, is what deals with the hardware device and the physical medium. This obviously covers items like hubs, repeaters, network adapters, etc. You can also place *cable specifications* and *pins* under this group.

The data layer deals mainly with the *procedural means* to transfer data between the network and the other elements in the network. This may perplex some people. Well, in order to avoid that confusion, you can assume that the physical layer simply meddles with the linking of a single entity with the physical medium, while the layer above it handles multiple devices. The data layer has the ability to perform error correction and control the flow of data.

The network layer deals with the transferring

of data sequences of different lengths from a given source to the intended destination. This layer does the routing functions and is also responsible for fragmentation and reassembly of data in many cases. The network layer can report delivery errors as well; and these facilitate the operation of routers at this layer. The Internet Protocol (IP) is the most well known among those belonging to this layer.

The transport layer is responsible for maintaining the quality of service. It does so by requesting this quality of the layer below it and, in turn, helping in reliable data transfer. The layer can also perform segmentation and de-segmentation processes. Just like the data layer, this layer plays a role in flow and error control as well. Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) are two examples that belong to this category.

The session layer, obviously, deals with the handling of sessions. It manages the connection between two computers (say, a local and remote application). The presentation layer is actually a link layer that does some sort of *translation* service. It takes inputs from the higher layer and feeds them into the session layer in such a way that the layer can handle them.

The top-most layer is the application layer that interacts with software applications and is architecturally positioned close to the end user. Typical examples could be Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Simple Mail Transfer Protocol (SMTP), etc.

In short, you can see that in the OSI model, the N^{th} -layer is supported by an $(N-1)^{\text{th}}$ layer, which helps the former layer to enable error-free transfer of data. These architectural changes were introduced as per the demand.

The Internet has expanded a lot over the last